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HQNet: A hybrid quantum network for multi-class MRI brain classification via quantum computing

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ABSTRACT

The complex and nonlinear nature of brain tumor morphology and texture pose significant challenges for representative feature extraction from magnetic resonance imaging (MRI) images, ultimately leading to inaccurate diagnoses of brain tumors. To tackle this problem, we propose a novel approach termed the hybrid quantum–classical convolutional network (HQNet) for detecting MRI brain tumors. Firstly, we use data augmentation techniques to deal with the issues of class imbalance and the sample scarcity within brain tumor MRI images. Secondly, we integrate channel attention and self-attention mechanisms to capture both channel and spatial dependencies from MRI images. This integration enables us to effectively fuse these dependencies and mitigate the discrepancies between them. However, the varying dependence of brain tumors at different scales can result in a decline in classification performance. To address this issue, we extract brain tumor features at various scales and combine them to alleviate this effect. Thirdly, a new parameterized quantum circuit is designed based on the principles of quantum computing to maximize the potential advantages of quantum computing and improve feature extraction. Finally, we compress the features and pass them through the fully connected layer to generate the final classification result. The effectiveness of our model is demonstrated through its superior performance when evaluated using three benchmark datasets.

1. Introduction

With the advancement of modern medicine, magnetic resonance imaging (MRI) has become one of the most commonly used neuroimaging techniques to detect and diagnose brain tumors since MRI scans provide precise neuroanatomical biomarkers (Martucci et al., 2023). MRI offers the advantages of safety and non-invasiveness. At the same time, it can obtain brain tumor images in different planes (cross-sectional, sagittal, coronal) to provide more comprehensive information. MRI technology can provide the location, size, shape, and other information of brain tumors for the diagnosis. Hospital experts can use MRI images of brain tumors for auxiliary diagnosis (Moayeri & Feizi, 2021). However, building a system that can accurately diagnose brain tumors is still a challenging task due to the various shapes, sizes, and locations of their growth (Shahin, Aly, & Aly, 2023).

Recently, there have been numerous applications in the field of deep learning (Baur, Albarqouni, & Navab, 2017; Chen, Wang, Gao & Zhou, 2018; Doulamis & Doulamis, 2014), particularly in brain tumor imaging. Yang et al. utilized a convolutional neural network (CNN) to categorize the ClinicalTrials.gov (NCT026226201) MRI brain tumor dataset, achieving an accuracy of 94.5% (Yang et al., 2018). Isunuri and Kakarla (2022) proposed a seven-layer CNN to address the brain tumor triple classification issue using separable convolutions. However, the reliability of these models is often compromised by issues like small sample sizes or imbalanced datasets (Xie et al., 2022). And these issues mitigated by employing data augmentation techniques (Gab Allah, Sarhan, & Elshennawy, 2021; Mzoughi et al., 2020). For example, Sajjad, Khan, Muhammad, Wu, Ullah, and Baik (2019) expanded dataset sizes using various augmentation methods, while Deepak and Ameer (2023) introduced class weighting to address the issue of imbalanced class distribution in brain tumor datasets. Evidently, data augmentation emerges as a promising technique, which can effectively deal with the case of small sample sizes and class imbalances in brain tumor classification. Motivated by this, we incorporate data augmentation into our work.

Additionally, attention mechanisms are widely utilized in various tasks (Liu, Miao, Fiumara, & De Meo, 2024), especially in image-related applications like image-based pollen detection (Li, Wang, Xiong, Zhao, Cheng, & Xu, 2024), facial expression recognition (Chen, Liang,

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Xue, & Yu, 2023), and computerized tomography (CT) image segmentation (Xuan et al., 2023). These mechanisms enable models to extract representative regional features within images. This benefit brought by the attention mechanism is also applied to the task of brain tumor classification. Saurav, Sharma, Saini, and Singh (2022) designed an AG-CNN to perform MRI brain tumor classification by incorporating channel attention blocks. Farajzadeh, Sadeghzadeh, and Hashemzadeh (2023) proposed a single-entity architecture method for brain tumor image segmentation and classification tasks by combining a hyper-kernel convolutional layer and a self-attention layer. Wang, Lu, Wang, and Zhang (2024) proposed the RanMerFormer network for brain tumor classification based on the traditional Transformer structure. Shahin et al. (2023) proposed a residual band pooling attention structure by combining the residual structure (He, Zhang, Ren, & Sun, 2016) and band pooling (Hou, Zhang, Cheng, & Feng, 2020), which is the key module of the MRI brain tumor classification framework MBTFCN. However, the attention mechanisms mentioned above only focus on channel information or context information, which may result in one-sided information. Therefore, we are inspired to propose a dual attention structure for aiding the feature extraction of brain tumor lesions. The proposed dual-attention structure combines channel information and the context dependence of brain tumors.

Breakthroughs in quantum computing technology have attracted significant attention from researchers (Arute et al., 2019). Quantum computing can perform certain computations more efficiently than classical computers. It can utilize the principles of quantum superposition and entanglement to simplify calculations (Rieffel & Polak, 2011), thereby reducing the complexity of deep learning methods.With the emergence of "Noisy Intermediate-Scale Quantum" (NISQ) computers, which offer stable computing power and can mitigate decoherence (Lü, Gao, Lü, Pan, & Dong, 2022; Preskill, 2018), researchers begin to explore the intersection of quantum computing and deep learning (Beer et al., 2020; Cerezo et al., 2021; Tian et al., 2022). As a result, many scholars have gradually attempted to explore hybrid neural networks based on quantum CNN (Doan, Sasdelli, Suter, & Chin, 2022; Konar et al., 2023; Liang, Peng, Zheng, Silvén, & Zhao, 2021).

In recent years, some success has been achieved by applying hybrid neural networks in the medical field (Wei et al., 2023). Mathur et al. (2022) utilized quantum neural networks to classify chest X-ray images. Essam et al. Houssein, Abohashima, Elhoseny, and Mohamed (2022) proposed an HQ-CNN model to predict COVID-19 by combining CNN and random quantum circuits. Azevedo, Silva, and Dutra (2022) combined variational quantum circuit and transfer learning to detect breast cancer. In the field of brain tumor classification, T, S, Manikumar, Dheeraj, and Sumanth (2022) proposed a CNN-quantum hybrid transfer learning method for classification. Amin, Anjum, Gul, and Sharif (2022) proposed a hybrid model combining a two-qubit quantum circuit and a fully connected layer for classification. Choudhuri and Halder (2023) utilized a parameterized quantum circuit comprising 16 qubits in conjunction with CNN to propose HQCNN for the binary classification of brain tumors. Javaria et al. Amin, Anjum, Gul, and Sharif (2023) combined the YOLOv2tiny model and the 7-layer J.Qnet model to locate and classify brain tumors. However, the existing parametric quantum circuits are relatively simple and may not fully leverage the advantages of quantum computing. On the other hand, overly complex quantum circuits could lead to a waste of computing resources. Therefore, we propose a Parameterized Quantum Circuit(PQC) that balances the cost and performance of quantum circuits.

To address the aforementioned challenges, we propose leveraging the intricate dependencies inherent in brain tumor MRI images, along with their nuanced variations across different scales. Our novel approach, termed the Hybrid Quantum-Classical Convolutional Network (HQNet), is designed to detect brain tumors effectively. We propose the Feature Extraction and Recombination (FER) module, which consists of six Self-proliferate Dual Attention (SPDA) modules and five Residual Spatial Pyramid Fusion Pooling (RSPFP) modules stacked on top of each other. In addition, in order to explore a broader parameter space and leverage quantum computing, we design a quantum circuit with 4 qubits, called Random Quantum Circuit (RQC) module. This design aims to balance the implementation cost and performance of quantum circuits by integrating the unique features of quantum superposition and entanglement. In summary, this paper's contributions are as follows:

- We designed and implemented a novel hybrid quantum neural network, HQNet, for MRI brain tumor detection. Harnessing the capabilities of quantum computing, HQNet enhances representative feature extraction, thereby augmenting the precision of tumor detection.
- We propose the SPDA module by combining the channel attention and self-attention mechanisms. This module can focus on channel information and context dependence in the image to highlight the characteristics of the tumor region.
- We propose the RSPFP module based on pyramid pooling and atrous spatial convolutional pyramid pooling. This module can mitigate the impact of dependencies at various scales.
- Based on existing quantum circuit connection methods and using the expressivity of quantum circuits as a benchmark, we propose the RQC module to strike a balance between cost and performance. This enables the exploration of a broader parameter space.

In order to test the performance of the proposed model, three public brain tumor datasets with different classification tasks were used. The experimental results show that our HQNet achieves 97.55% accuracy on the BT-large-4c dataset, which outperforms other existing methods. In addition, HQNet achieves 98.96% accuracy on the Cheng dataset and 99% accuracy on the BT-large-2c dataset for various classification tasks. This demonstrates that the model is well-suited for different brain tumor classification tasks.

2. Proposed method

The complete brain tumor classification process is divided into two parts: training and testing. The execution flow of the entire system is illustrated in Fig. 1. Firstly, we use operations such as translation and rotation to augment the dataset. Subsequently, the dataset is divided into a training set and a test set for model training and performance evaluation, respectively. Next, the training set is used to obtain the trained model. Finally, the test set is used to evaluate this model based on the classification results. During the training phase, the images are sequentially passed through the four modules (Hybrid Residual Module, FER Module, RQC Module, Classification Module) of HQNet. For the Classification(CL) module, it consists of a global average pooling layer and two fully connected layers. The global average pooling stretches the feature map after the quantum convolutional layer into one dimension and uses it as the input for the fully connected layer. The first fully connected layer uses the ReLU activation function, and the second fully connected layer uses the softmax function for multi-classification tasks. The structure of the HQNet model is depicted in Fig. 2.

2.1. Hybrid residual module

The Hybrid Residual (HR) module consists of one stem convolutional block and two residual modules, as shown in Fig. 3(a). The stem convolutional block consists of one convolutional layer, one batch normalization layer, and one ReLU activation function. The HR module utilizes two residual blocks with different kernel sizes. Res_A employs a 5×5 convolution kernel on the main branch to expand the receptive field, while Res_B utilizes a 3×3 kernel to enhance generalization ability, followed by a ReLU activation function. In each residual block,

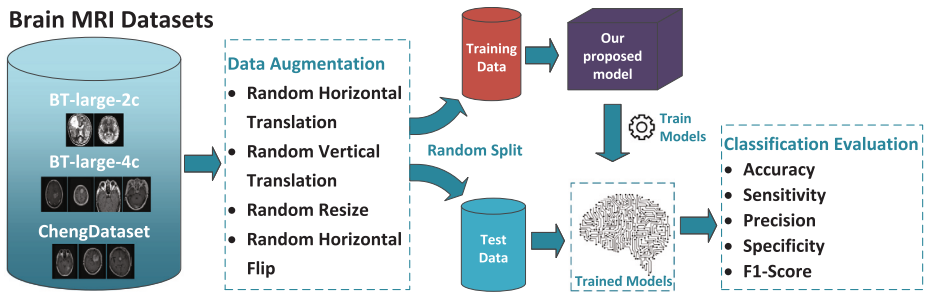


Fig. 1. Complete workflow of brain tumor classification model.

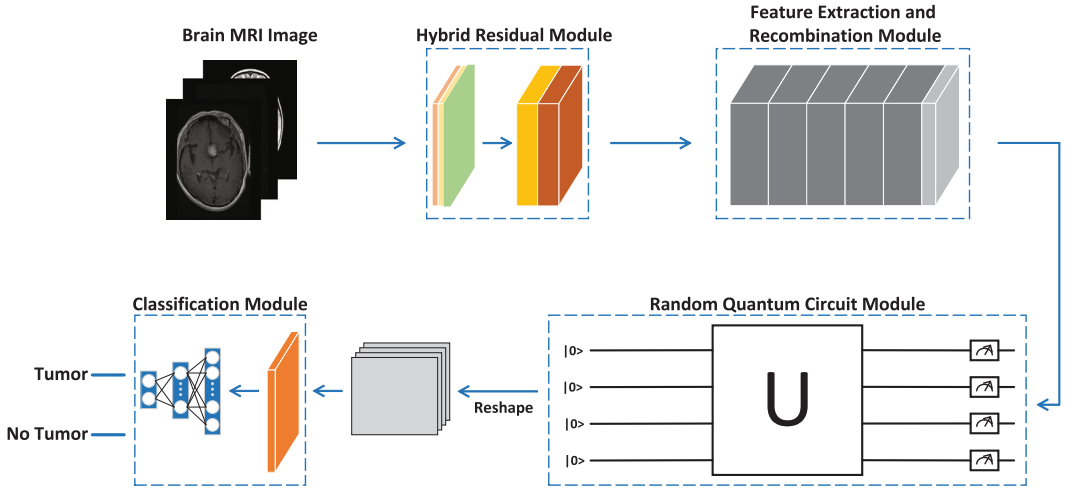


Fig. 2. The proposed network for brain tumor classification.

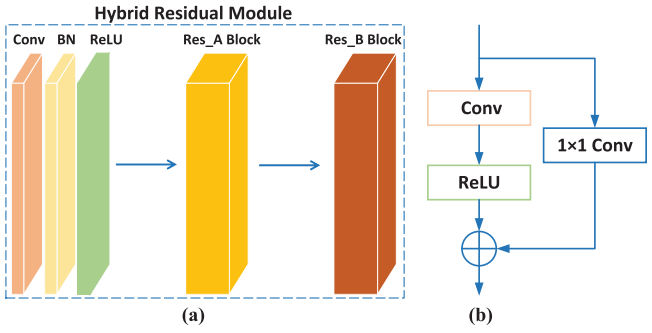


Fig. 3. (a) The proposed Hybrid Residual module, (b) structure of residual block.

there is a 1 × 1 convolutional layer on the shortcut branch to adjust to the change in the number of channels on the main branch. The specific structure is shown in Fig. 3(b). The input image passes through the Stem convolution block, Res_A block, and Res_B block sequentially.

2.2. Feature extraction and recombination module

The FER module consists of cross-stacked SPDA modules and RSPFP modules (Chen, Papandreou, Kokkinos, Murphy, & Yuille, 2018). The overall structure of the SPDA module is shown in Fig. 4(a). The self-proliferation method (Yang & Sun, 2022) was introduced, and the Channel and Self-Attention Block (CSAB) was constructed by combining the channel attention mechanism and self-attention mechanism. The specific structure of CSAB is shown in Fig. 4(b). The self-proliferation method first requires the ordinary convolution operation to establish the mapping of the features to obtain the new feature map, as shown in Eq. (1). Then, the depth-wise convolution operation is used to linearly transform the features obtained in (1), doubling the number of feature maps, as shown in Eq. (2).

$$M_{new} = M_{old} * filter \tag{1}$$

Where $M_{new} \in \mathbb{R}^{h' \times w' \times out_c}$, $M_{old} \in \mathbb{R}^{h \times w \times in_c}$, $filters \in \mathbb{R}^{k \times k \times out_c}$, $*$ represents the convolution operation, h,w(h′, w′) represents the height and width of the original feature map (new feature map), k represents the size of the convolution kernel, and in_c(out_c) represents the input (output) channel.

$$\rho_{i,j} = LT(m'_i), \forall i = 1, \ldots, s, j = 1, \ldots, t \tag{2}$$

Where m'_i is the ith unit of M_{new} generated by Eq. (1), and LT is a linear transformation.

The CSAB module consists of two branches: one branch represents the channel attention structure, which utilizes the Squeeze-and-Excitation (SE) structure (Hu, Shen, & Sun, 2018). The SE block is a classical module for modeling channel dependencies. It exhibits strong transferability and can be integrated with other network structures, all while maintaining low computational costs. The other branch represents the self-attention structure, and its design is modeled after the Global Context (GC) block (Cao, Xu, Lin, Wei, & Hu, 2019). The infrastructure of the GC block can be expressed as Eq. (3).

$$\mathbf{z}_i = \mathbf{x}_i + W_{v2}\mathrm{ReLU}\left(\mathrm{LN}\left(W_{v1}\sum_{j=1}^{N_p}\frac{e^{W_k\mathbf{x}_j}}{\sum_{m=1}^{N_p}e^{W_k\mathbf{x}_m}}\mathbf{x}_j\right)\right) \tag{3}$$

Where $\frac{e^{W_k\mathbf{x}_j}}{\sum_{m=1}^{N_p}e^{W_k\mathbf{x}_m}}$ represents global attention pooling and $W_{v2}\mathrm{ReLULN}(W_{v1}(\cdot))$ represents the bottleneck structure. Compared with the GC block, the bottleneck structure suitable for deep networks is omitted in our self-attention structure branch to reduce the amount of model calculation. Instead, a 1 × 1 convolution is used to adapt to the

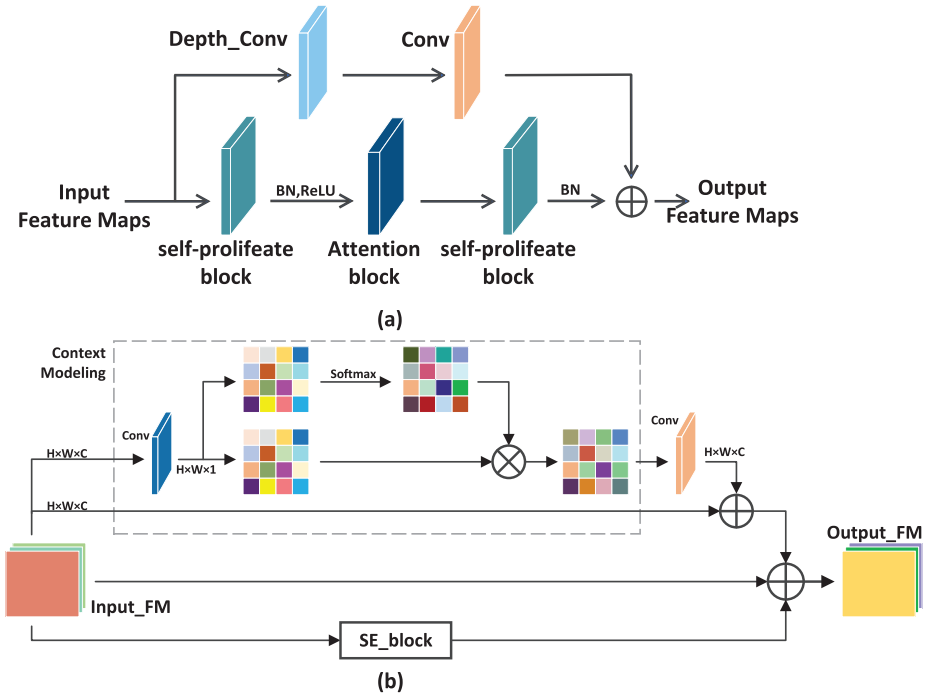


Fig. 4. (a) Structure of SPDA module, (b) structure of CSAB.

number of channels, followed by feature fusion through element-by-element addition, as illustrated in the upper part of Fig. 4(b). Therefore, through Equation (3), the proposed self-attention branch structure can be represented by Eq. (4).

$$\mathbf{b}_i = \mathbf{x}_i + W_{v2}\left(\sum_{j=1}^{N_p} \frac{e^{W_k \mathbf{x}_j}}{\sum_{m=1}^{N_p} e^{W_k \mathbf{x}_m}} \mathbf{x}_j\right) \tag{4}$$

Due to the difference in morphology and location of brain tumors within the same category, it is necessary to fuse feature information at different scales. The HQNet model utilizes multiple pooling methods for feature fusion at various scales.

RSPFP utilizes two pooling methods: average pooling and max pooling, which are employed to sample the input feature maps, at two different scales: 2×2 and 3×3. After each pooling module, a dilated convolutional layer, regularization layer, and ReLU activation function are applied. Each convolution kernel has the same size but different atrous rates, which correspond to different scales. Feature fusion is then performed on the feature maps that have been pooled at the same scale. It is worth noting that there is a blank branch and a Max pooling branch in RSPFP. The empty branch is designed to preserve the original features, while the other branch is intended to extract features from the feature map without feature fusion. The RSPFP module is designed in this manner to thoroughly extract regional and morphological information from brain tumors in various ways, while also retaining the existing features as much as possible. After the aforementioned steps, the feature maps produced by the various methods are reorganized and then passed through the subsequent CBR(Conv+BN+ReLU) and max pooling with a step size of 2.

2.3. Random quantum circuit module

In the RQC module, the quantum circuit is divided into three parts: the encoding layer, the entanglement layer, and the measurement layer. Classical data cannot be directly applied to quantum circuits. Therefore, there is a coding layer whose function is to transform classical data into quantum states. There are generally three ways to convert classical data into quantum states: basic coding, amplitude coding, and angle coding (Yang & Sun, 2022). Similarly, quantum states cannot be directly applied to classical data. Therefore, the function of the measurement layer is to convert quantum states into classical states. At the same time, the measurement layer also performs a crucial nonlinear function.

Angle coding is utilized in our encoding layer to facilitate the transformation of data from classical states to quantum states. Suppose there are N qubits and initialize the quantum state to $|0\rangle$; R denotes the single-qubit rotation gate and x_k denotes the classical information, the conversion process from a classical state to a quantum state can be formulated as

$$|x_k\rangle = \otimes_{k=0}^{N} R(x_k)|0^N\rangle \tag{5}$$

Where R denotes the two kinds of rotation gates R_x and R_z, which are represented by the matrices of rotations around the x-axis and z-axis of the Bloch sphere, respectively. The Pauli spin matrices represented by R_x and R_z are as follows.

$$R_x(x_k) = \begin{bmatrix} \cos\frac{x_k}{2} & -i\sin\frac{x_k}{2} \\ -i\sin\frac{x_k}{2} & \cos\frac{x_k}{2} \end{bmatrix} \tag{6}$$

$$R_z(x_k) = \begin{bmatrix} 1 & 0 \\ 0 & e^{ix_k} \end{bmatrix} \tag{7}$$

In addition, CR_x, CR_z two types of controlled revolving doors are also used in this study, whose matrices are shown in Eqs. (8) and (9). As shown in Fig. 5(c), CR_x gate requires two qubits, the control bit is represented by a black dot, and the controlled bit is represented by the R_x gate. The quantum circuit proceeds from left to right, similar to the classical circuit. The representation of the four types of quantum gates R_x, R_z, CR_x, and CR_z in the quantum circuit is shown in Fig. 5.

$$CR_x(x_k) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & cos\frac{(x_k)}{2} & -isin\frac{(x_k)}{2} \\ 0 & 0 & -isin\frac{(x_k)}{2} & cos\frac{(x_k)}{2} \end{bmatrix} \tag{8}$$

$$CR_z(x_k) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & cos\frac{(x_k)}{2} + isin\frac{(x_k)}{2} \end{bmatrix} \tag{9}$$

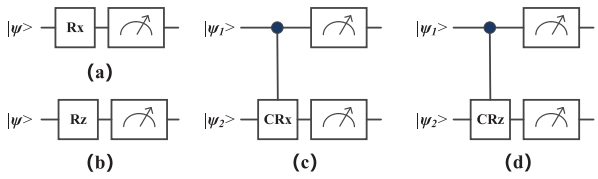


Fig. 5. Circuit representation of Rx, Rz, CRx, CRz.

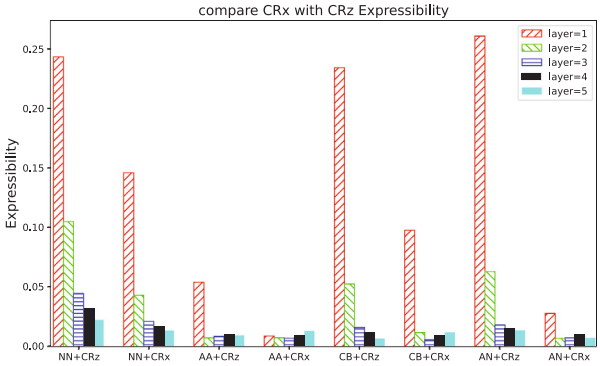


Fig. 6. The results of the expressibility comparison.

The entanglement layer contains two-qubit rotating quantum gates, and the rotation angle of these quantum gates can be used as parameters for training to find the best position in the Hilbert space. We refer to Hilbert space through expressiveness, which is defined as the circuit's ability to generate pure states (Sim, Johnson, & Aspuru-Guzik, 2019). It is calculated by determining the Kullback–Leibler (KL) divergence between the state samples collected from the PQC and the Haar distribution, as shown in Eq. (10).

$$\mathrm{Expr} = D_{\mathrm{KL}}(\hat{P}_{\mathrm{PQC}}(F;\boldsymbol{\theta}) \parallel P_{\mathrm{Haar}}(F)) \tag{10}$$

Where $\hat{P}_{\mathrm{PQC}}(F;\boldsymbol{\theta})$ is the estimated probability distribution of the fidelity produced by the sampled states of the PQC, θ is the parameter on the PQC, and $P_{\mathrm{Haar}}(F)$ can be expressed as follows.

$$P_{\mathrm{Haar}}(F) = (N-1)(1-F)^{N-2} \tag{11}$$

Where F denotes the fidelity and N denotes the dimension of the Hilbert space.

We selected the nearest-neighbor (NN), circuit-block (CB), all-to-all (AA), and the proposed adjacent connection (AC) to compare the expressiveness of CR_x and CR_z. We respectively set CR_x and CR_z on the same type of circuit, and each circuit type will undergo a control experiment with circuit depths ranging from 1 layer to 5 layers. The experimental results are shown in Fig. 6.

It can be seen from the results in Fig. 6 that when the circuit depth is relatively shallow, the expressiveness of the CR_x rotation gate is generally better than that of the CR_z rotation gate. Meanwhile, we find that the expressiveness of the circuit is also influenced by the different connection methods of the circuit when the quantum gate is fixed. For example, the expressiveness of AC lies between CB and AA, but the cost of AC is lower than that of AA. Another point worth noting is that as the number of layers increases, the expressiveness of each circuit will become more similar, but this comes at the expense of a significant increase in the cost of implementing the quantum circuit. In this case, the NN connection with the minimum cost should be adopted. However, it is more realistic to use the shallow quantum circuit model. Therefore, the depth of the proposed quantum circuit is 1, and the proposed adjacent connection method (AC+CR_x) is utilized. The specific circuit is shown in Fig. 7.

3. Experiments

3.1. Experimental setup

The framework used to implement this model is TensorFlow 2.9.3 and CUDA 11. The GPU specification is 24 GB, and the CPU specification is 24 GB with 8 cores. The experiments are carried out on three public brain tumor datasets, and the detailed information is presented in Table 1.

As depicted in Table 1, the three datasets offer distinct perspectives on brain anatomy, and each one encompasses varying numbers of brain tumor classes from patients with diverse tumor grades and ages. The dataset follows a common partition strategy, with 70% allocated for training, 10% for validation, and 20% for testing. In the process of network training, we use the Adam optimizer with β_1 and β_2 set to 0.9 and 0.999, respectively. The number of training rounds is 30 on the BT-large-4c dataset and 20 on the other two datasets. Every 5 rounds, the learning rate is reduced by multiplying it by 0.5. We set the minimum batch size to 32 and the L2 regularization parameter to 0.0001. The hyperparameter values during training were set as shown in Table 2.

The metrics to evaluate the experimental results include accuracy, recall, precision, specificity, and F1-Score. Their definitions are follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{12}$$

$$Recall = \frac{TP}{TP+FN} \tag{13}$$

$$Specificity = \frac{TN}{FP+TN} \tag{14}$$

$$Precision = \frac{TP}{TP+FP} \tag{15}$$

$$F1\text{-}Score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{16}$$

Where True Positive (TP) represents the number of brain tumor cases correctly predicted as positive. True Negative (TN) is the number of cases predicted as negative that is negative. False Positive (FP) is the number of cases that were incorrectly predicted to be negative but were actually positive for a brain tumor. False Negative (FN) represents the number of cases that were incorrectly predicted to be positive but were actually negative for a brain tumor.

3.2. Experimental results

In order to evaluate the performance of our proposed model, we conducted experiments on three different public datasets corresponding to three classification tasks. Meanwhile, to evaluate the overall performance of the entire model, we calculate the average value of each index. The detailed experimental results are shown in the following sections.

3.2.1. Performance evaluation on the BT-large-4c dataset

Fig. 8 and Table 3 correspond to the classification and evaluation of the BT-large-4c dataset. The dataset comprises a total of 653 test images, with 30 training rounds conducted, resulting in an overall accuracy of 97.55%. It can be seen that the model has the best classification effect for pituitary tumors, with accuracy and specificity both at 100%. The recall rate is 98.92%, and the F1-Score is as high as 99.46%. This may be because the shape and size characteristics of pituitary tumors are more apparent. The model achieved the highest recall rate of 99.01% for the tumor-free class. Additionally, it demonstrated specificity and F1-Score evaluation results of 99.10% and 97.09%, respectively. However, the precision is relatively low at 95.24%. From the perspective of the corresponding confusion matrix, the primary reason for the low accuracy of the tumor-free class was that the model misclassified four glioma images as tumor-free. Similarly, the model

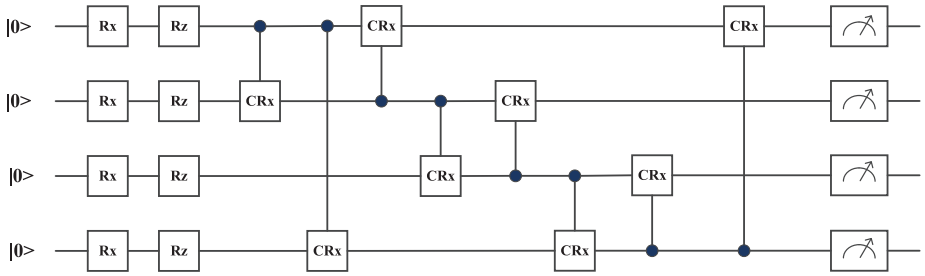


Fig. 7. Structure of Random Quantum Circuit.

Table 1
Data sets used in the experiments.

No.	Dataset name	Categories of Brain Tumors	The number of samples in each class	Total
1	BT-large-2c	Tumor	1500	3000
		Non-tumor	1500	
2	BT-large-4c	Glioma	826	3264
		Meningioma	822	
		Pituitary	827	
		No tumor	395	
3	Cheng Dataset	Glioma	1426	3064
		Meningioma	708	
		Pituitary	930	

Table 2
Setting of hyperparameter values for the proposed network.

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	0.001
β_1	0.9
β_2	0.999
epoch	30(20)
Learn decay rate	0.5/5 epoch
Minimum batch size	32
L2 regularization parameter	0.0001

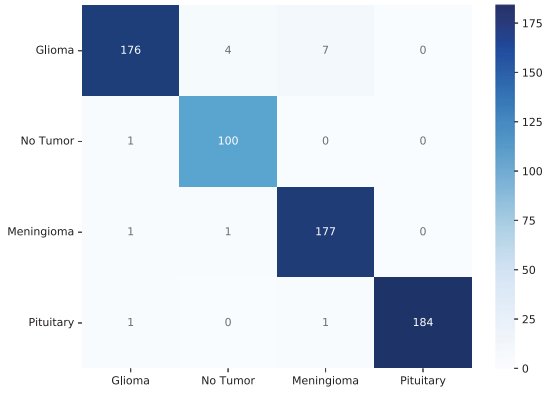


Fig. 8. The confusion matrix of the BT-large-4c.

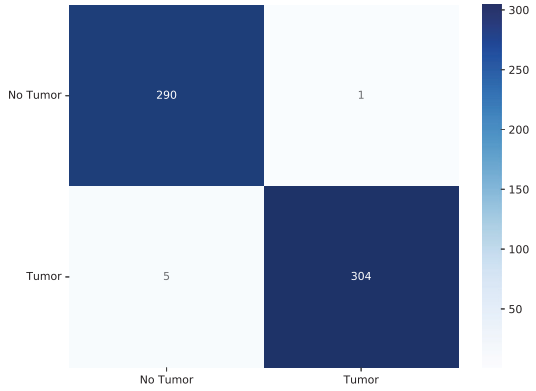


Fig. 9. The confusion matrix of the BT-large-2c.

Table 3
BT-large-4c classification evaluation.

Category	Precision	Recall	Specificity	F1-score	Support
Glioma	98.32%	94.18%	99.36%	96.17%	187
No Tumor	95.24%	**99.01%**	99.10%	97.09%	101
Meningioma	95.68%	98.88%	98.31%	97.25%	179
Pituitary	**100%**	98.92%	**100%**	**99.46%**	186
Avg./Total	97.31%	97.73%	99.19%	97.49%	653

Table 4
BT-large-2c classification evaluation.

Category	Precision	Recall	Specificity	F1-score	Support
No Tumor	98.31%	**99.66%**	98.38%	98.98%	291
Tumor	**99.67%**	98.39%	**99.66%**	**99.02%**	309
Avg./Total	98.99%	99.02%	99.02%	99.00%	600

misclassified 7 images of glioma as meningioma, resulting in a precision of 95.68%. The recall rate for glioma was only 94.18% due to the incorrect classification of glioma into other categories. The reason for this phenomenon may be that glioma and meningioma have high morphological similarity in some images.

3.2.2. Performance evaluation on the BT-large-2c dataset

Fig. 9 and Table 4 display the results of the binary classification task conducted by the model on the BT-large-2c dataset. The training consists of 20 rounds, focusing solely on determining the presence of a tumor. According to the experimental results, we can see that the overall accuracy of this task is 99.00%. It has the highest recall f 99.66% for the No Tumor class, with precision and F1-Score evaluation results of 98.31% and 98.98%, respectively. For the Tumor class, it has the highest precision of 99.67%, but the recall rate is relatively low at 98.39% because the model misclassifies five tumor images as

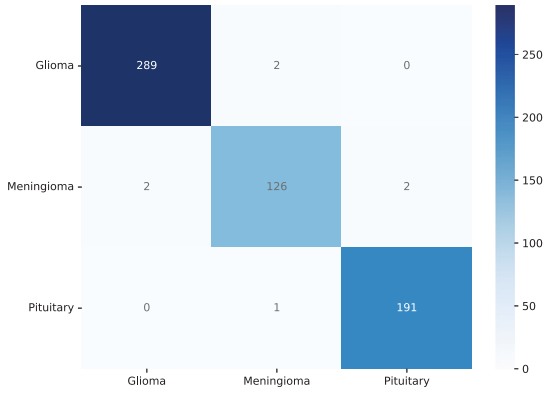


Fig. 10. The confusion matrix of the Cheng Dataset.

Table 5
Cheng dataset classification evaluation.

Category	Precision	Recall	Specificity	F1-score	Support
Glioma	**99.31%**	99.31%	99.38%	**99.31%**	291
Meningioma	97.67%	96.92%	99.38%	97.30%	130
Pituitary	98.96%	**99.48%**	**99.52%**	99.22%	192
Avg./Total	98.65%	98.57%	99.43%	98.61%	683

Table 6
Proposed models used in the ablation study.

Module	Accuracy	F1-Score	Recall	Specificity	Precision
HR-SPDA-CL	0.9311	0.9304	0.9287	0.9767	0.9326
HR-RSPFP-CL	0.9602	0.9612	0.9636	0.9863	0.9594
HR-FER-CL	0.9693	0.9705	0.9730	0.9895	0.9691
HR-FER-RQC-CL	0.9755	0.9753	0.9775	0.9900	0.9731

non-tumor. In general, the average value of each secondary index is hovering around 99% (±0.02%). It can be seen that our proposed model has a very high correct diagnosis rate for the binary classification task of brain tumors. This may be attributed to the fact that the model only needs to determine whether it belongs to a brain tumor, rather than identifying the specific type of brain tumor.

3.2.3. Performance evaluation on the cheng dataset

The Cheng dataset corresponds to a three-class classification task. Unlike the previous two tasks, this task does not involve the tumor-free classification. According to Fig. 10 and Table 5, the overall accuracy of this task is 98.86%. The recall rate of our model for pituitary tumors is 99.48%, with only one image being misrecognized as meningioma. And its F1-Score index is 99.62%. The classification indices of glioma are relatively similar, with precision, recall, and F1-Score indices at 99.31%, representing thll rate of meningioma was 96.92% due to four images being mistakenly classified as other types of brain tumors, given the relatively small sample size. In general, the average precision, recall, and F1-Score of our model are 98.65%, 98.57%, and 98.61%, respectively. The model still maintains excellent classification performance even when the dataset is imbalanced.

3.2.4. The ablation study

In order to assess the impact of subtle differences in various types of brain tumors on the model and to evaluate the functionality of each module within our model, we conducted a series of ablation studies using the BT-large-4c dataset, which contains the largest variety and volume of brain tumor data. As depicted in Table 8, we performed a total of four experiments. The HR-SPDA-CL model excluded the RSPFP module, while the HR-RSPFP-CL model excluded the SPDA module. Additionally, the HR-FER-CL model integrated both SPDA and RSPFP modules, whereas the HR-FER-RQC-CL model incorporated an additional RQC module onto the base HR-FER-CL model.

Table 7
Comparison of network models on the BT-large-4c dataset.

Model	Accuracy	F1-Score	Precision	Recall	Specificity
ResNet50 (He et al., 2016)	94.03%	93.83%	93.73%	94.05%	98.01%
DenseNet121 (Huang et al., 2017)	95.71%	95.60%	95.27%	96.10%	98.59%
InceptionV3 (Szegedy et al., 2016)	95.10%	95.27%	95.24%	95.45%	98.34%
MobileNet (Howard et al., 2017)	95.10%	95.14%	94.75%	95.63%	98.37%
Xception (Chollet, 2017)	96.02%	95.91%	95.74%	96.14%	98.67%
EfficientNetB0 (Tan & Le, 2019)	96.48%	96.59%	96.54%	96.65%	98.80%
HQNet	**97.55%**	**97.49%**	**97.31%**	**97.73%**	**99.19%**

Table 8
Comparison of network model parameters and memory.

Model	#Para (M)	Size (MB)
ResNet50	25.64	97.81
DenseNet121	8.07	30.77
InceptionV3	23.86	91.00
MobileNet	4.26	16.24
Xception	22.91	87.41
EfficientB0	5.33	20.35
HQNet	**2.75**	**10.48**

Based on Table 6, it is clear that the RSPFP module plays a significant role. The HR-SPDA-CL model without the RSPFP module has the lowest accuracy at only 93.11%. Meanwhile, the HR-RSPFP-CL model without the SPDA module achieves an accuracy of 96.02%, and when both modules are combined in the HR-FER-CL model, the accuracy increases to 96.93%. This indicates that the FER module formed by combining the SPDA and RSPFP modules can exert maximum influence on the model's performance. Furthermore, it is apparent that the impact of the RSPFP module outweighs that of the SPDA module.This suggests that feature extraction from different scales is more effective in capturing subtle shape and size differences between brain tumors. Additionally, integrating the RQC module into the HR-FER-CL model leads to the highest accuracy rate of 97.55%, demonstrating the positive impact of incorporating parameterized quantum circuits to extract high-dimensional features that are difficult for classical networks to capture.

3.3. Comparison with baselines

We compare existing models (Chollet, 2017; He et al., 2016; Howard, Zhu, Chen, Kalenichenko, Wang, Weyand, Andreetto, & Adam, 2017; Huang, Liu, Van Der Maaten, & Weinberger, 2017; Szegedy, Vanhoucke, Ioffe, Shlens, & Wojna, 2016; Tan & Le, 2019) including ResNet50, DenseNet121, InceptionV3, MobileNet, Xception, and EfficientNetB0, with our proposed HQNet model in terms of classification performance. The experiments are carried out on the BT-large-4c dataset. Except for the HQNet model, the input size of the other six models is 224, which does not affect the total number of parameters in the model. And it can also best reflect the performance of the model. For a fair comparison, all layers of the pre-trained CNN model are fine-tuned with the same settings to achieve better performance. The experimental results are shown in Table 7, and Table 8 provides a comparison of parameters and the required memory space.

We can see that compared with our proposed model, ResNet50 and Xception networks require more parameters and memory space, achieve 94.03% and 96.02% accuracy, respectively. Meanwhile, our proposed model, HQNet, achieves a higher accuracy of 97.55% and requires only 2.75 million parameters and 10.48MB of memory space. It is easy to found that compared with such as MobileNet, EfficientNetB0, DenseNet121, and InceptionV3, our model also demonstrates superior performance. This validates its exceptional feature learning ability when working with limited training data. Fig. 11 illustrates the performance comparing our proposed HQNet model with various current popular CNN models. From the Fig. 11, we can clearly see that our model has fewer parameters and higher accuracy. The model with more parameters, such as ResNet50, is not necessarily good.

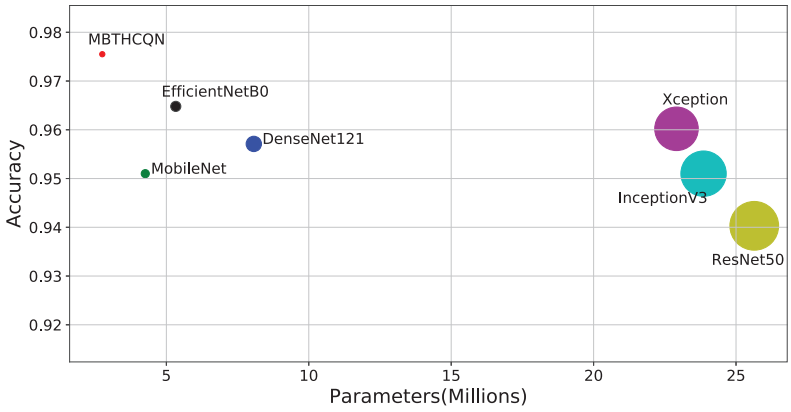


Fig. 11. Accuracy versus model size (parameters) for various CNN architectures.

Table 9
Compares with the existing methods.

Model	Dataset	Accuracy	F1-Score	Precsion	Recall	Specificity
DenseNet-169+Shufflenet +MnasNet (Kang, Ullah, & Gwak, 2021)	BT-large -4c	0.9158	no detail	no detail	no detail	no detail
AG-CNN (Saurav et al., 2022)		0.9571	0.9598	0.9577	0.9633	0.9852
MBTFCN (Shahin et al., 2023)		0.9632	0.9630	0.961	0.9654	0.9877
HQNet		**0.9755**	**0.9749**	**0.9731**	**0.9773**	**0.9919**
DenseNet-121+ResNeXt -101+MnasNet (Kang et al., 2021)	BT-large -2c	0.9883	no detail	no detail	no detail	no detail
Shape-based features+ ensemble classifier (Bansal & Jindal, 2022)		0.9892	0.9892	0.9892	0.9892	0.9892
HQNet		**0.9900**	**0.9900**	**0.9899**	**0.9902**	**0.9902**
Shape-based features+ ensemble classifier (Bansal & Jindal, 2022)	Cheng Dataset	0.9848	0.9747	0.9763	0.9735	0.9882
AG-CNN (Saurav et al., 2022)		0.9723	0.9700	0.9707	0.9695	0.9857
HQNet		**0.9886**	**0.9861**	**0.9865**	**0.9857**	**0.9943**

4. Comparison and discussion with existing advanced brain tumor classification methods

In order to evaluate the progress of our proposed model, the results are presented in Table 9, comparing them with recent studies while maintaining the same method of dividing the dataset. We compared all three datasets with the current existing results.

It can be seen that our designed HQNet achieves superior results on the three datasets. Kang et al. (2021) combined various pre-trained neural network models and achieved 91.58% and 98.83% accuracy on the BT-large-4c and BT-large-2c datasets, but did not provide more detailed evaluation indicators to explain these results. Sumeet Saurav (Saurav et al., 2022) achieved 95.71% and 97.23% accuracy on BT-large-4c and Cheng datasets by utilizing a CNN with a self-designed channel attention structure, and our method increases by 1.84% and 1.63% respectively. Compared with the residual strip pooling attention method of Shahin et al. (2023), our proposed HQNet method has a better effect and shows an improvement of 1.23%. Bansal and Jindal (2022) extracted mixed features based on brightness, texture, and shape and used different classifiers to achieve 98.92% and 98.48% accuracy on BT-large-2c and Cheng datasets.

In addition, it can be observed that for the BT-large-4c dataset, which involves more complex classification tasks, the diagnostic rate of existing models will be significantly reduced. Although our network also experienced a slight decrease, it has the smallest decrease compared to other models, highlighting one of the aspects of its superior performance.

It can also be seen from the Table 9 that research on brain tumor classification in the past two years primarily utilized deep learning methods, particularly the attention mechanism and multi-scale feature extraction method. This validates the rationality and effectiveness of our proposed FER module.

5. Conclusions and future work

In this paper, a novel hybrid quantum circuit neural network (HQNet) is proposed, which consists of four modules: HR module, FER module, RQC module, and CL module. The FER module includes the SPDA module and the RSPFP module. By cross-stacking, the SPDA module can more fully extract the dependencies in MRI brain tumor images. Meanwhile, the RSPFP module is also able to detect subtle differences in dependencies at various scales. In addition, we integrate parametric quantum circuits into CNN, aiming to leverage the benefits of quantum computing and explore a broader parameter space. And our quantum circuits enhance the performance of the entire brain tumor classification network. Compared with the previous quantum–classical hybrid neural network in the field of MRI brain tumor classification, our network model enables to tackle more complex tasks. Our proposed method achieves the highest accuracy on the BT-large-4c dataset when compared with the current popular CNN architectures and recent methods.

The current limitation is that there are almost no researches about MRI brain tumor images segmentation, which may make the model's suitability for complex brain tumor classification tasks and upgrading its classification ability. In future research, we will focus on designing a more robust model that combines deep learning and parametric quantum circuits for.

CRediT authorship contribution statement

Aijuan Wang: Supervision. **Dun Mao:** Conceptualization, Methodology. **Xiangqi Li:** Formal analysis, Resources. **Tiehu Li:** Writing – review & editing. **Lusi Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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